The intervals of concavity are tabulated in Table 7.7.

Interval	$(-\infty, 1)$	$(1, 3)$	$(3, \infty)$
Sign of $f''(x)$	+	−	+
Concavity	concave up	concave down	concave up

Table 7.7

The curve is concave upwards on $(-\infty,1)$ and $(3,\infty)$.

The curve is concave downwards on $(1,3)$.

As $f''(x)$ changes its sign when it passes through $x=1$ and $x=3$, $(1, f(1))=(1,0)$ and $(3, f(3))=(3,-16)$ are points of inflection for the graph $y=f(x)$. The sign change may be observed from the adjoining figure of the curve $f''(x)$.

Example 7.58

Determine the intervals of concavity of the curve $y=3+\sin x$.

Solution

The given function is a periodic function with period 2π and hence there will be stationary points and points of inflections in each period interval. We have,

$$\frac{dy}{dx} = \cos x \text{ and } \frac{d^2y}{dx^2} = -\sin x$$

$$\text{Now, } \frac{d^2y}{dx^2} = -\sin x = 0 \Rightarrow x = n\pi .$$

We now consider an interval, $(-\pi,\pi)$ by splitting into two sub intervals $(-\pi,0)$ and $(0,\pi)$.

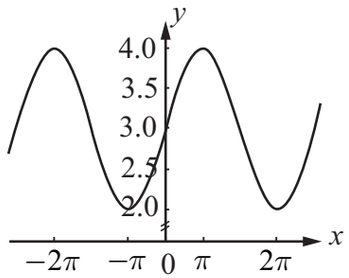


Fig.7.25

In the interval $(-\pi,0)$, $\frac{d^2y}{dx^2}>0$ and hence the function is concave upward.

In the interval $(0,\pi)$, $\frac{d^2y}{dx^2}<0$ and hence the function is concave downward. Therefore $(0,3)$ is a point of inflection (see Fig. 7.25). The general intervals need to be considered to discuss the concavity of the curve are $(n\pi,(n+1)\pi)$, where n is any integer which can be discussed as before to conclude that $(n\pi,3)$ are also points of inflection.

7.7.2 Extrema using Second Derivative Test

The Second Derivative Test: The Second Derivative Test relates the concepts of critical points, extreme values, and concavity to give a very useful tool for determining whether a critical point on the graph of a function is a relative minimum or maximum.

Theorem 7.13 (The Second Derivative Test)

Suppose that c is a critical point at which $f'(c)=0$, that $f'(x)$ exists in a neighborhood of c, and that $f''(c)$ exists. Then f has a relative maximum value at c if $f''(c)<0$ and a relative minimum value at c if $f''(c)>0$. If $f''(c)=0$, the test is not informative.

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